Low-field carrier mobility models for 3D electron gas (3DEG) slab

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This documentation is part of `mobilitypy` package.[1]

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Following models for low-field electron mobility (μ) calculations from different scattering mechanism are based on the Reference-[2, 3, 4, 5, 6, 7] and the references therein.

The available mobility model versions are:

- 'v1' (available in `mobilitypy` $\geq$ `v1.0.5`)

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1. Following Ref.[4], we calculate bulk mobility (total mobility as a function of position):

$$\mu(z) = \left(\sum_i \frac{1}{\mu_i(z)} \right)^{-1}$$

   where μ_i's are the mobility from different scattering mechanisms presented below.

2. Following Ref.[4], the effective mobility (effective scattering rate $\propto \frac{1}{\mu_{eff}}$) is calculated as the first moment of the carrier density ($n_{3d}(z)$) w.r.t the local scattering rate ($\propto \frac{1}{\mu(z)}$):

$$\frac{1}{\mu_{eff}} = \frac{\int \frac{n_{3d}(z)}{\mu(z)}\, dz}{\int n_{3d}(z)\, dz} \implies \mu_{eff} = \frac{\int n_{3d}(z)\, dz}{\int \frac{n_{3d}(z)}{\mu(z)}\, dz}$$

3. We also output the effective mobility calculated as the first moment of the carrier density ($n_{3d}(z)$) w.r.t the local carrier mobility:

$$\implies \mu_{eff} = \frac{\int n_{3d}(z)\mu(z)\, dz}{\int n_{3d}(z)\, dz}$$

1 Parameters

Parameters name	Symbol	SI Unit
Electron elementary charge	e	C
Boltzmann constant	k_B	$\mathrm{J\,K^{-1}}$
Temperature	T	K
Mass density	ρ	$\mathrm{kg\,m^{-3}}$
Lattice parameter	a_0	Å
Lattice parameter	c_0	Å
Unit cell volume	Ω_0	$\mathrm{Å}^3$
Bandgap energy	E_g	eV
Alloy-disordered scattering potential	U_0	eV
Electron effective mass	m^*	m_0
Electron rest mass	m_0	kg
Static dielectric constant	ϵ_s	ϵ_0
High frequency dielectric constant	ϵ_h	ϵ_0
Vacuum permittivity	ϵ_0	$\mathrm{C\,V^{-1}\,m^{-1}}$
Longitudinal acoustic phonon velocity	v_{LA}	$\mathrm{m\,s^{-1}}$
Transversal acoustic phonon velocity	v_{TA}	$\mathrm{m\,s^{-1}}$
Conduction band deformation potential	E_D	eV
Polar optical phonon energy	E_{POP}	eV
Electromechanical coupling coefficient	K^2	-
Elastic constants	C_{ij}	GPa
Fermi wave vector	k_F	$10^6\ \mathrm{cm^{-1}}$
Thomas-Fermi wave vector	q_{TF}	$10^6\ \mathrm{cm^{-1}}$
POP wave vector	q_0	$10^6\ \mathrm{cm^{-1}}$
Volumetric electron gas density	n_{3d}	$10^{18}\ \mathrm{cm^{-3}}$
Dislocation density	N_{DIS}	$10^8\ \mathrm{cm^{-2}}$
Dislocation occupancy	f_{DIS}	-
Mobility	μ	$\mathrm{cm^{-2}\,V^{-1}\,s^{-1}}$

Note: Proper unit conversions for all parameters have been implemented within the equations.

2 Basic theory of mobility

- **Fermi-Dirac function for fermions:**

$$f_0(E) = \frac{1}{1 + e^{\frac{E-E_F}{k_B T}}} \tag{1}$$

where E_F is the Fermi energy.

- **Boltzmann Transport Equation (BTE)**: Within relaxation time approximation:

$$f(\vec{k}) = f_0(\vec{k}) + e\mathbb{F}_i \tau(k) v_i \frac{\partial f_0}{\partial \varepsilon} \tag{2}$$

where $f(\vec{k})$ is carrier distribution function, $\tau(k)$ is the momentum relaxation time, v_i is the carriers velocity in the i-th direction in reponse of the field $\mathbb{F}_i$.

- **DOS in d-dimension:**

$$g_d(E) = \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \left(\frac{2m^*}{\pi\hbar^2}\right)^{\frac{d}{2}} (E - E_C)^{\frac{d}{2}-1} \tag{3}$$

- **Total number of carriers/unit 'volume' in d-dimensional space:**

$$\begin{aligned}
n_{dd} &= \int \frac{d^d k}{(2\pi)^d} f(k) \\
&= \int_{E_C}^{\infty} f_0(E)\, g_d(E)\, dE \\
&= \int_{E_C}^{\infty} \frac{1}{1 + e^{(E-E_F)/k_B T}}\, g_d(E) dE \\
&= \int_{E_C}^{\infty} \frac{1}{1 + e^{(E-E_F)/k_B T}} \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \left(\frac{2m^*}{\pi\hbar^2}\right)^{\frac{d}{2}} (E - E_C)^{\frac{d}{2}-1} dE \\
&= \left(\frac{2m^*}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \int_{E_C}^{\infty} \frac{1}{1 + e^{(E-E_F)/k_B T}} (E - E_C)^{\frac{d}{2}-1} dE \\
&= \left(\frac{2m^*}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \int_{E_C}^{\infty} \frac{1}{1 + e^{\frac{(E-E_C)-(E_F-E_C)}{k_B T}}} \left(\frac{E - E_C}{k_B T}\right)^{\frac{d}{2}-1} (k_B T)^{\frac{d}{2}-1}\, dE \\
&= \left(\frac{2m^* k_B T}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}\Gamma(\frac{d}{2})} \int_{E_C}^{\infty} \frac{1}{1 + e^{\frac{(E-E_C)-(E_F-E_C)}{k_B T}}} \left(\frac{E - E_C}{k_B T}\right)^{\frac{d}{2}-1} \frac{1}{k_B T}\, dE
\end{aligned} \tag{4}$$

- **Fermi-Dirac integral of j-th order:**

$$F_j(\zeta) = \frac{1}{\Gamma(j+1)} \int_0^{\infty} \frac{u^j}{1 + e^{u-\zeta}}\, du \tag{5}$$

- Substituting $\varepsilon = \frac{E-E_C}{k_B T}$ and $\eta_F = \frac{E_F - E_C}{k_b T}$

$$f_0(\varepsilon) = \frac{1}{1 + e^{\varepsilon - \eta_F}} \tag{6}$$

$$F_{\frac{d}{2}-1}(\eta_F) = \frac{1}{\Gamma(\frac{d}{2})} \int_0^{\infty} \frac{\varepsilon^{\frac{d}{2}-1}}{1 + e^{\varepsilon-\eta_F}}\, d\varepsilon = \frac{1}{\Gamma(\frac{d}{2})} \int_0^{\infty} f_0(\varepsilon)\, \varepsilon^{\frac{d}{2}-1}\, d\varepsilon \tag{7}$$

$$\begin{aligned}
n_{dd} &= \left(\frac{2m^* k_B T}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}} \frac{1}{\Gamma(\frac{d}{2})} \int_0^{\infty} \frac{1}{1 + e^{\varepsilon-\eta_F}} \varepsilon^{\frac{d}{2}-1}\, d\varepsilon \\
&= \left(\frac{2m^* k_B T}{\pi\hbar^2}\right)^{\frac{d}{2}} \frac{1}{2^{d-1}} F_{\frac{d}{2}-1}(\eta_F)
\end{aligned} \tag{8}$$

- **Current:**

$$J_i = en \underbrace{\left(-\frac{2e}{dm^*} \frac{\int_{E_C}^{\infty} \tau_m (E-E_C)^{d/2} \frac{\partial f_0}{\partial E} dE}{\int_{E_C}^{\infty} f_0 \, (E-E_C)^{\frac{d}{2}-1} \, dE} \right)}_{\mu_{dd}} \mathcal{F}_i \tag{9}$$

- **Mobility in d-dimension:**

$$\begin{aligned}
\mu_{dd} &= -\frac{2e}{dm^*} \frac{\int_{E_C}^{\infty} \tau_m (E-E_C)^{d/2} \frac{\partial f_0(E)}{\partial E} dE}{\int_{E_C}^{\infty} f_0(E) \times (E-E_C)^{\frac{d}{2}-1} \, dE} && (10)\\
&= -\frac{2e}{dm^*} \frac{\int_{E_C}^{\infty} \tau_m (E-E_C)^{d/2} \frac{\partial f_0(E)}{\partial E} \, dE}{\int_{E_C}^{\infty} f_0(E) \times (E-E_C)^{\frac{d}{2}-1} \, dE} \\
&= -\frac{2e}{dm^*} \frac{(k_B T)^{\frac{d}{2}} \int_0^{\infty} \tau_m \varepsilon^{d/2} \, \frac{\partial f_0(\varepsilon)}{\partial \varepsilon} d\varepsilon}{(k_B T)^{\frac{d}{2}} \int_0^{\infty} f_0 \times \varepsilon^{\frac{d}{2}-1} \, d\varepsilon} \\
&= -\frac{e\tau_0}{m^*} \frac{\int_0^{\infty} \varepsilon^{\frac{d}{2}+p} \, \frac{\partial f_0(\varepsilon)}{\partial \varepsilon} d\varepsilon}{\frac{d}{2} \int_0^{\infty} f_0 \times \varepsilon^{\frac{d}{2}-1} \, d\varepsilon} \; [\because \tau_m = \tau_0 \times \varepsilon^p] && (11)\\
&= -\frac{e\tau_0}{m^*} \frac{-\Gamma(\frac{d}{2}+p+1) \, F_{\frac{d}{2}+p-1}(\eta_F)}{\frac{d}{2} \times \Gamma(\frac{d}{2}) \times F_{\frac{d}{2}-1}(\eta_F)} \\
&= \frac{e\tau_0}{m^*} \left(\frac{\Gamma(\frac{d}{2}+p+1)}{\Gamma(\frac{d}{2}+1)} \right) \left(\frac{F_{\frac{d}{2}+p-1}(\eta_F)}{F_{\frac{d}{2}-1}(\eta_F)} \right) && (12)
\end{aligned}$$

Since, $f_0(\varepsilon) \to 0$ as $\varepsilon \to \infty$ and $\varepsilon^m \to 0$ as $\varepsilon \to 0$, using integration by parts $[\int f(x) g'(x) dx = f(x)g(x) - \int f'(x) g(x) dx]$

$$\begin{aligned}
& \int_0^{\infty} \varepsilon^{\frac{d}{2}+p} \frac{\partial f_0(\varepsilon)}{\partial \varepsilon} \, d\varepsilon \\
&= [\varepsilon^{\frac{d}{2}+p} \, f_0(\varepsilon)]_0^{\infty} - \int_0^{\infty} \left(\frac{d}{2}+p \right) \times \varepsilon^{\frac{d}{2}+p-1} \, f_0(\varepsilon) \, d\varepsilon \\
&= -\left(\frac{d}{2}+p \right) \int_0^{\infty} \varepsilon^{\frac{d}{2}+p-1} \, f_0(\varepsilon) \, d\varepsilon \\
&= -\left(\frac{d}{2}+p \right) \Gamma \left(\frac{d}{2}+p \right) \, F_{\frac{d}{2}+p-1}(\eta_F) \\
&= -\Gamma \left(\frac{d}{2}+p+1 \right) \, F_{\frac{d}{2}+p-1}(\eta_F) && (13)
\end{aligned}$$

- **In 3D:**

$$\begin{aligned}
n_{3d} &= \left(\frac{2m^* \, k_B T}{\pi \hbar^2} \right)^{\frac{3}{2}} \frac{1}{4} \, F_{\frac{1}{2}}(\eta_F) \\
&= 2 \left(\frac{m^* \, k_B T}{2 \, \pi \hbar^2} \right)^{\frac{3}{2}} \, F_{\frac{1}{2}}(\eta_F) \\
&= N_C^{3d} \, F_{\frac{1}{2}}(\eta_F) && (14)\\
N_C^{3d} &= 2 \left(\frac{m^* \, k_B T}{2 \, \pi \hbar^2} \right)^{\frac{3}{2}} && (15)\\
\mu_{3d} &= \frac{e\tau_0}{m^*} \frac{\Gamma(\frac{5}{2}+p)}{\Gamma(\frac{5}{2})} \frac{F_{\frac{1}{2}+p}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)} \\
&= \frac{4e\tau_0}{3\sqrt{\pi} \, m^*} \frac{1}{F_{\frac{1}{2}}(\eta_F)} \Gamma \left(\frac{5}{2}+p \right) \times F_{\frac{1}{2}+p}(\eta_F) && (16)
\end{aligned}$$

3 Fermi-Dirac integral

3.1 Inverse Fermi-Dirac integral of order 1/2

The scaled Fermi energy ($\eta_F = \frac{E_F - E_C}{k_B T}$) is calculated using inverse Fermi-Dirac integral of order 1/2.

$$\begin{aligned} \because n_{3d} &= N_C^{3d}\, F_{\frac{1}{2}}(\eta_F) \,\Big|\, = N_C^{3d}\, \frac{2}{\sqrt{\pi}} \mathcal{F}_{\frac{1}{2}}(\eta_F) = N_{3d}\, \mathcal{F}_{\frac{1}{2}}(\eta_F) \\ \implies \eta_F &= F_{1/2}^{-1}\left(\frac{n_{3d}}{N_C^{3d}}\right) \,\Big|\, \eta_F = \mathcal{F}_{1/2}^{-1}\left(\frac{n_{3d}}{N_{3d}}\right) \end{aligned} \tag{17}$$

where

$$F_{\frac{1}{2}}(\eta_F) = \frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\varepsilon^{\frac{1}{2}}}{1 + e^{\varepsilon - \eta_F}} d\varepsilon \,\Big|\, \mathcal{F}_{\frac{1}{2}}(\eta_F) = \int_0^\infty \frac{\varepsilon^{\frac{1}{2}}}{1 + e^{\varepsilon - \eta_F}} d\varepsilon \tag{18}$$

$$N_C^{3d} = 2\left(\frac{m^*\, k_B T}{2\,\pi \hbar^2}\right)^{\frac{3}{2}} \Big|\, N_{3d} = \frac{2}{\sqrt{\pi}}\, N_C^{3d} = \frac{4}{\sqrt{\pi}} \left(\frac{m^*\, k_B T}{2\,\pi \hbar^2}\right)^{\frac{3}{2}} \tag{19}$$

1. **Using Joyce-Dixon approximation:**[8]

$$\begin{aligned} \eta_F &= F_{1/2}^{-1}\left(\frac{n_{3d}}{N_C^{3d}}\right) \simeq \ln\left(\frac{n_{3d}}{N_C^{3d}}\right) + \sum_{l=1}^{3} A_l \times \left(\frac{n_{3d}}{N_C^{3d}}\right)^l \\ A_l &= 0.3535534, -4.95009 \times 10^{-3}, 1.48386 \times 10^{-4} \end{aligned} \tag{20}$$

   Note that the equation here is in terms of n_{3d} and N_C^{3d}. To compare with the original derivation in Ref.[8] use this notation:
   $[\mathfrak{F} = 2\pi^{-1/2}\, F]_{publication} \equiv [F_{1/2} = 2\pi^{-1/2}\, \mathcal{F}_{1/2}]_{our}$

2. **Using minimax rational function approximation:**[9]

$$\eta_F = \mathcal{F}_{1/2}^{-1}\left(\frac{n_{3d}}{N_{3d}}\right) \simeq \text{Piecewise function [Ref.[9]]} \tag{21}$$

   Note that the equation here is in terms of n_{3d} and N_{3d}.

   Unit analysis (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned} N_C^{3d} &= 2\left(\frac{m^*\, k_B T}{2\pi\hbar^2}\right)^{3/2} \\ &= 2\left(\frac{k_B\, m_0}{2\pi\hbar^2}\right)^{3/2} \times 10^{-6}\,\text{cm}^{-3}\text{K}^{-3/2} \times (m^{*'}\, T')^{3/2}\,\text{K}^{3/2} \\ &= 4.83 \times (m^{*'}\, T')^{3/2} \times 10^{15}\ \text{cm}^{-3} \end{aligned} \tag{22}$$

$$N_{3d} = \frac{2}{\sqrt{\pi}} N_C^{3d} = \frac{2}{\sqrt{\pi}} \times 4.83 \times (m^{*'}\, T')^{3/2} \times 10^{15}\ \text{cm}^{-3} \tag{23}$$

$$= 5.45 \times (m^{*'}\, T')^{3/2} \times 10^{15}\ \text{cm}^{-3} \tag{24}$$

$$\frac{n_{3d}}{N_C^{3d}} = \frac{n'_{3d} \times 10^{18}}{4.83 \times (m^{*'}\, T')^{3/2} \times 10^{15}} = 207.07 \times \frac{n'_{3d}}{(m^{*'}\, T')^{3/2}} \tag{25}$$

$$\frac{n_{3d}}{N_{3d}} = \frac{n'_{3d} \times 10^{18}}{5.45 \times (m^{*'}\, T')^{3/2} \times 10^{15}} = 183.51 \times \frac{n'_{3d}}{(m^{*'}\, T')^{3/2}} \tag{26}$$

3.2 Fermi-Dirac integrals

1. η_F is calculated using expression from Sec.3.1.

2. **Fermi-Dirac integral of order 0: (analytical solution)**

$$F_0(\eta_F) = \frac{1}{\Gamma(1)} \int_0^\infty \frac{1}{1+e^{\varepsilon-\eta_F}} d\varepsilon = \ln[1+\exp(\eta_F)]$$

3. **Fermi-Dirac integral of order 1/2:**

$$F_{\frac{1}{2}}(\eta_F) = \frac{1}{\Gamma(\frac{1}{2}+1)} \int_0^\infty \frac{\varepsilon^{\frac{1}{2}}}{1+e^{\varepsilon-\eta_F}} d\varepsilon = \frac{2}{\sqrt{\pi}} \int_0^\infty \frac{\varepsilon^{\frac{1}{2}}}{1+e^{\varepsilon-\eta_F}} d\varepsilon$$

4. **Fermi-Dirac integral of order 1:**

$$\begin{aligned}
F_1(\eta_F) &= \frac{1}{\Gamma(2)} \int_0^\infty \frac{\varepsilon}{1+e^{\varepsilon-\eta_F}} d\varepsilon = \int_0^\infty \frac{\varepsilon}{1+e^{\varepsilon-\eta_F}} d\varepsilon && (27)\\
F_1(\eta_F) &= -\mathrm{Li}_2(-e^{\eta_F}) && (28)\\
\mathrm{Li}_2(z) &= -\int_0^z \frac{\ln(1-t)}{t} dt = \texttt{scipy.spence(1-z)} \\
\implies \mathrm{Li}_2(-e^{\eta_F}) &= \texttt{scipy.spence(1+e^eta_f)} \\
\implies F_1(\eta_F) &= (-1) \times \texttt{scipy.spence(1+e^eta_f)} && (29)
\end{aligned}$$

The integration can be calculated numerically using Eqn. 27 (implemented using `scipy.quad`), or using dilogarithm formalism of complete Fermi-Dirac integral of order-1, Eqn. 28 (implemented as `(-1) * scipy.spence(1+e^eta_f)`), or using Fukushima's piecewise minimax rational approximation[10]. All are implemented in our package.

4 Carrier degeneracy limits

Generalized formula for *j-th order* Fermi-Dirac integral:

$$F_j(\eta) = \frac{1}{\Gamma(j+1)} \int_0^\infty \frac{x^j}{1+e^{x-\eta}} dx \tag{30}$$

Generalized formula for carrier density in *d-dimensional* space:

$$n_{dd} = N_C^{dd} F_{\frac{d}{2}-1}(\eta) \tag{31}$$

Generalized mobility formula for *arbitrary* degeneracy carriers in *d-dimension*:

$$\mu_{dd} = \frac{e\tau_0}{m^*} \left(\frac{\Gamma(\frac{d}{2}+p+1)}{\Gamma(\frac{d}{2}+1)} \right) \left(\frac{F_{\frac{d}{2}+p-1}(\eta)}{F_{\frac{d}{2}-1}(\eta)} \right) \tag{32}$$

4.1 Strongly non-degenerate limit ($\eta << -1$):

$$\begin{aligned}
F_j^{ND}(\eta) &\approx e^\eta \mid F_{\frac{1}{2}}^{ND}(\eta) \approx e^\eta \\
n_{dd}^{ND} &\approx N_C^{dd} e^\eta \;\Big|\; n_{3d}^{ND} \approx N_C^{3d} e^\eta && (33)\\
\mu_{dd}^{ND} &\approx \frac{e\tau_0}{m^*} \left(\frac{\Gamma(\frac{d}{2}+p+1)}{\Gamma(\frac{d}{2}+1)} \right) \;\Big|\; \mu_{3d}^{ND} \approx \frac{e\tau_0}{m^*} \left(\frac{\Gamma(\frac{5}{2}+p)}{\Gamma(\frac{5}{2})} \right) && (34)
\end{aligned}$$

4.2 Strongly degenerate (metallic) limit ($\eta >> 1$):

Transport occurs at the Fermi level.

$$F_j^D(\eta) \approx \frac{\eta_F^{j+1}}{\Gamma(j+2)} \Bigg| \; F_{\frac{1}{2}}^D(\eta) \approx \frac{\eta_F^{\frac{3}{2}}}{\Gamma(\frac{5}{2})} \tag{35}$$

$$\mu_{dd}^D \approx \frac{e\tau_0}{m^*}\eta_F^p \;\Big|\; \mu_{3d}^D \approx \frac{e\tau_0}{m^*}\eta_F^p \tag{36}$$

$$n_{dd}^D \approx N_C^{dd}\,\frac{\eta_F^{\frac{d}{2}}}{\Gamma(\frac{d}{2}+1)} \Bigg| \; n_{3d}^D \approx N_C^{3d}\,\frac{\eta_F^{\frac{3}{2}}}{\Gamma(\frac{5}{2})} \tag{37}$$

$$\eta_F = \frac{E_F}{k_B T} \Leftrightarrow E_F = \frac{\hbar^2\,k_F^2}{2\,m^*} \Leftrightarrow k_F = (3\,\pi^2\,n_{3d})^{\frac{1}{3}} \tag{38}$$

$$\Longrightarrow N_C^{3d}\,\eta^{\frac{3}{2}} = 2\left(\frac{m^*\,k_B T}{2\,\pi\,\hbar^2}\right)^{\frac{3}{2}} \frac{E_F^{\frac{3}{2}}}{(k_B T)^{\frac{3}{2}}} = 2\left(\frac{m^*\,E_F}{2\,\pi\,\hbar^2}\right)^{\frac{3}{2}} \tag{39}$$

NOTE-1: Carrier density depends exponentially with temperature for *non-degenerate* case. The temperature dependence cancel for *degenerate* case, indicating metallic behavior.
NOTE-2: The validity of the limits depends on the accuracy of the approximations made to the Fermi-Dirac integrals.

5 Scattering mechanisms

$$k_F = \left(3\pi^2 n_{3d}\right)^{\frac{1}{3}} \tag{40}$$

$$\lambda_{TF} = q_{TF}^{-1} = \sqrt{\frac{2\epsilon\varepsilon_F}{3e^2 n_{3d}}} = \sqrt{\frac{2\epsilon}{3e^2 n_{3d}}\frac{\hbar^2 k_F^2}{2\,m^*}} = \frac{\hbar k_F}{e}\sqrt{\frac{\epsilon}{3n_{3d}m^*}}$$

$$\varepsilon_F = \frac{\hbar^2 k_F^2}{2\,m^*} \tag{41}$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$k_F = \left(3\pi^2 n_{3d}\right)^{\frac{1}{3}} = 3.094 \times n_{3d}'^{\frac{1}{3}}\;[10^6\,\mathrm{cm}^{-1}] \tag{42}$$

$$\lambda_{TF} = \frac{\hbar\left(3\pi^2 n_{3d}\right)^{\frac{1}{3}}}{e}\sqrt{\frac{\epsilon_s}{3n_{3d}m^*}} = \frac{\hbar\,\pi^{\frac{2}{3}}\epsilon_0^{\frac{1}{2}}}{3^{\frac{1}{6}}\,e\,m_0^{\frac{1}{2}}} \times \left(\frac{\epsilon_s}{n_{3d}^{\frac{1}{3}}\,m'^*}\right)^{\frac{1}{2}}$$

$$= 0.037 \times \left(\frac{\epsilon_s}{n_{3d}^{\frac{1}{3}}\,m^{*\prime}}\right)^{\frac{1}{2}}\;[10^{-6}\,\mathrm{cm}] \tag{43}$$

$$\varepsilon_F = \frac{\hbar^2 k_F^2}{2\,m^*} = \frac{\hbar^2 \times (10^6)^2 \times 10^4}{2\,m_0 \times e'}\frac{k_F'^2}{m'^*} = 0.00038 \times \frac{k_F'^2}{m'^*}\;[\mathrm{eV}] \tag{44}$$

5.1 Structural related scattering

5.1.1 Alloy-disorder scattering

The implementation is based on Refs.[3, 4]. For alloy disorder mediated scattering $p = -\frac{1}{2}$. The **momentum relaxation time:**

$$\tau_{0,AD} = \frac{\hbar}{2\sqrt{\pi}}\frac{k_B\,T}{U_0^2\,\Omega_0\,x\,(1-x)\,N_C^{3d}} \tag{45}$$

Mobility expression:

$$\begin{aligned}
\mu_{3d,AD} &= \frac{4e\,\tau_{0,AD}}{3\sqrt{\pi}\,m^*}\frac{1}{F_{\frac{1}{2}}(\eta_F)}\Gamma(2)\times F_0(\eta_F)\ \text{[From Eqn.16]}\\
&= \frac{2e\hbar}{3\pi}\frac{k_B\,T}{m^*\,U_0^2\,\Omega_0\,x\,(1-x)}\frac{1}{N_C^{3d}\,F_{\frac{1}{2}}(\eta_F)}F_0(\eta_F)\\
&= \frac{2e\hbar}{3\pi}\frac{k_B\,T}{m^*\,U_0^2\,\Omega_0\,x\,(1-x)\,n_{3d}}F_0(\eta_F)\ \text{[From Eqn.14]} \qquad (46)
\end{aligned}$$

Re-writing as function position, z:

$$\mu_{3d,AD}(z) = \frac{2e\hbar}{3\pi}\frac{k_B\,T}{m^*(z)\,U_0(z)^2\,\Omega_0(z)\,x(z)\,(1-x(z))\,n_{3d}(z)}F_0(\eta_F(z)) \qquad (47)$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\mu_{3d,AD}(z) &= 21.17\times\frac{T'}{m^{*\prime}(z)\,U_0'(z)^2\,\Omega_0'(z)\,x(z)\,(1-x(z))\,n_{3d}'(z)}\times F_0(\eta_F(z))\\
&\qquad [\text{cm}^2\,\text{V}^{-1}\,\text{s}^{-1}] \qquad (48)
\end{aligned}$$

5.1.2 Dislocation scattering

5.1.2.1 Scattering from line charge of threading dislocation core

- Dislocation line charge density, $\rho_{disl.} = ef_{disl.}/c_0$. e is the electron charge, c_0 is the lattice constant along the dislocation axis, f is the fractional occupancy.

5.1.2.2 General scattering rate/mobility expression

The derivation is adapted from Ref.[11, 12].
Momentum relaxation scattering rate:

$$\begin{aligned}
\frac{1}{\tau^c_{disl.}(k)} &= \frac{n_{disl.}\,m^*\,e^2\,\rho^2_{disl.}}{\epsilon_s^2\,\hbar^3\,q_s^4\left(1+\frac{4k_\perp^2}{q_s^2}\right)^{\frac{3}{2}}}\\
&= \frac{n_{disl.}\,m^*\,e^4\,f^2_{disl.}}{\epsilon_s^2\,\hbar^3\,c_0^2}\frac{1}{q_s^4\left(1+\frac{4k^2}{q_s^2}\right)^{\frac{3}{2}}} \quad [\because \rho_{disl.} = ef/c_0] \qquad (49)
\end{aligned}$$

• For simplicity, the threading dislocations are assumed to be vertical, i.e., $k_\perp = k\cdot\sin\theta \approx k$ [$\theta = 90°$].

$$\begin{aligned}
\Longrightarrow \frac{1}{\tau^c_{disl.}(E_k)} &= \frac{n_{disl.}\,m^*\,e^4\,f^2_{disl.}}{\epsilon_s^2\,\hbar^3\,c_0^2}\frac{1}{q_s^4\left(1+\frac{8m^*E_k}{\hbar^2q_s^2}\right)^{\frac{3}{2}}} \quad \left[\because E_k = \frac{\hbar^2k^2}{2m^*}\right]\\
&= \frac{n_{disl.}\,m^*\,e^4\,f^2_{disl.}}{(8m^*)^{\frac{3}{2}}\epsilon_s^2\,c_0^2}\frac{1}{q_s\left(\frac{\hbar^2q_s^2}{8m^*}+E_k\right)^{\frac{3}{2}}} \qquad (50)\\
\Longrightarrow \tau^c_{disl.}(\varepsilon) &= \frac{16\sqrt{2}\,m^{*\frac{1}{2}}\epsilon_s^2\,c_0^2\,(k_BT)^{\frac{3}{2}}\,q_s}{n_{disl.}\,e^4\,f^2_{disl.}}\left(\frac{\hbar^2q_s^2}{8m^*\,k_BT}+\frac{E_k}{k_BT}\right)^{\frac{3}{2}} \qquad (51)\\
&\equiv A(B+\varepsilon)^{\frac{3}{2}} \qquad (52)
\end{aligned}$$

where $q_s = \sqrt{\frac{e^2N_c^{3d}F_{-\frac{1}{2}}(\eta_F)}{\epsilon_s\,k_BT}} = \sqrt{\frac{e^2n_{3d}}{\epsilon_s\,k_BT}}\sqrt{\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)}}$; $\varepsilon = \frac{E_k-E_c}{k_BT}$; $\eta_F = \frac{E_F-E_c}{k_BT}$;

$B = \frac{\hbar^2q_s^2}{8m^*\,k_BT} = \frac{\hbar^2e^2n_{3d}}{8m^*\epsilon_s\,(k_BT)^2}\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)}$; $A = \frac{16\sqrt{2}\,m^{*\frac{1}{2}}\epsilon_s^2\,c_0^2\,q_s\,(k_BT)^{\frac{3}{2}}}{n_{disl.}\,e^4\,f^2_{disl.}} = \frac{16\sqrt{2}\,m^{*\frac{1}{2}}\epsilon_s^{\frac{3}{2}}\,c_0^2k_BT\,n_{3d}^{\frac{1}{2}}}{n_{disl.}\,e^3\,f^2_{disl.}}\sqrt{\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)}}$

Mobility expression:

$$\mu^{3d}_{disl.} = -\frac{2e}{3m^*}\frac{\int_0^\infty \tau_c\,\varepsilon^{3/2}\,\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon}{\int_0^\infty f_0\times\varepsilon^{\frac{1}{2}}\,d\varepsilon} \tag{53}$$

$$= -\frac{2e}{3m^*}\frac{\int_0^\infty \tau_c\,\varepsilon^{3/2}\,\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon}{\Gamma(\frac{3}{2})\times F_{\frac{1}{2}}(\eta_F)}$$

$$= -\frac{4}{3\sqrt{\pi}}\frac{e}{m^*}\frac{\int_0^\infty \tau_c\,\varepsilon^{3/2}\,\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon}{F_{\frac{1}{2}}(\eta_F)}\left[\equiv -\frac{e}{m^*}\frac{\int_0^\infty \tau_c\,\varepsilon^{3/2}\,\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon}{\Gamma(\frac{5}{2})\times F_{\frac{1}{2}}(\eta_F)}\right] \tag{54}$$

$$= \frac{4}{3\sqrt{\pi}}\frac{e}{m^*}\frac{1}{F_{\frac{1}{2}}(\eta_F)}\frac{3A\sqrt{B}}{2}\times I(\eta_F)$$

$$= \frac{8\sqrt{2}}{\pi^2}\frac{m^{*\frac{1}{2}}\epsilon_s\,c_0^2\,(k_BT)^{\frac{3}{2}}}{\hbar^2 e n_{disl.} f^2_{disl.}}\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)}\times I(\eta_F) \tag{55}$$

Where $I(\eta_F) = \int_0^\infty \frac{(B+2\varepsilon)\left(1+\frac{\varepsilon}{B}\right)^{\frac{1}{2}}\varepsilon^{\frac{1}{2}}}{1+e^{\varepsilon-\eta_F}}\,d\varepsilon$.

$$\int_0^\infty \tau_c\,\varepsilon^{3/2}\,\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon = \int_0^\infty A\times(B+\varepsilon)^{\frac{3}{2}}\times\varepsilon^{\frac{3}{2}}\times\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon$$

$$= A\left[(B\varepsilon+\varepsilon^2)^{\frac{3}{2}}f_0\Big|_0^\infty - \int_0^\infty \frac{\partial}{\partial\varepsilon}\left[(B\varepsilon+\varepsilon^2)^{\frac{3}{2}}\right]f_0\,d\varepsilon\right] \tag{56}$$

$$= -\frac{3A}{2}\int_0^\infty (B\varepsilon+\varepsilon^2)^{\frac{1}{2}}(B+2\varepsilon)f_0\,d\varepsilon$$

$$= -\frac{3A\sqrt{B}}{2}\int_0^\infty \frac{(B+2\varepsilon)\left(1+\frac{\varepsilon}{B}\right)^{\frac{1}{2}}\varepsilon^{\frac{1}{2}}}{1+e^{\varepsilon-\eta_F}}\,d\varepsilon \equiv -\frac{3A\sqrt{B}}{2}\times I(\eta_F) \tag{57}$$

Integration by parts: $\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx$ is used to arrive at Eqn. 56. Limits are $f_0(\varepsilon)\to 0$ as $\varepsilon\to\infty$ and $\varepsilon^m\to 0$ as $\varepsilon\to 0$.
Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$B = \frac{\hbar^2 e^2 n_{3d}}{8m^*\epsilon_s\,(k_BT)^2}\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)} = 23210.2\times\frac{n'_{3d}}{m^{*\prime}\epsilon'_s T'^2}\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)} \tag{58}$$

$$\mu^{3d}_{disl.} = \frac{8\sqrt{2}}{\pi^2}\frac{m^{*\frac{1}{2}}\epsilon_s\,c_0^2\,(k_BT)^{\frac{3}{2}}}{\hbar^2 e n_{disl.} f^2_{disl.}}\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)}\times I(\eta_F)$$

$$= 0.0279\times\frac{c_0'^2\epsilon'_s}{n'_{disl.}\,f'^2_{disl.}}\sqrt{m^{*\prime}\,T'^3}\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)}\times I(\eta_F)\ \ [\mathrm{cm^2\,V^{-1}\,s^{-1}}] \tag{59}$$

$$I(\eta_F) = \int_0^\infty \frac{(B+2\varepsilon)\left(1+\frac{\varepsilon}{B}\right)^{\frac{1}{2}}\varepsilon^{\frac{1}{2}}}{1+e^{\varepsilon-\eta_F}}\,d\varepsilon \tag{60}$$

- η_F is calculated using inverse Fermi-Dirac function (i.e., $\eta_F = F^{-1}_{1/2}(\frac{n_{3d}}{N_C^{3d}})$).
- $F_{-\frac{1}{2}}(\eta_F)$, $F_{\frac{1}{2}}(\eta_F)$ and $I(\eta_F)$ are then calculated using numerical integration.

5.1.2.3 Degenerate limit

The implementation is based on Refs.[11, 5, 7, 6]. Note that, momentum relaxation rate in this case can not be written as polynomial function of energy. The **momentum relaxation rate:**

$$\frac{1}{\tau^{3d}_{disl.}} = \frac{m^* e^4 n_{disl.} f^2_{disl.}}{c_0^2 \epsilon_s^2 \hbar^3} \times \frac{\lambda^4_{TF}}{(1+4\,k_F^2\lambda^2_{TF})^{3/2}} \tag{61}$$

$$= \frac{m^* e^4 n_{disl.} f^2_{disl.}}{c_0^2 \epsilon_s^2 \hbar^3} \times \frac{\hbar^4 k_F^4 \epsilon_s^2}{9\,e^4\,n_{3d}^2 m^{*2}} \times \frac{1}{(1+4\,k_F^2\lambda^2_{TF})^{3/2}}$$

$$= \frac{n_{disl.} f^2_{disl.} \hbar}{m^* c_0^2} \times \frac{\pi^{8/3}}{(3\,n_{3d})^{2/3}} \times \frac{1}{(1+4\,k_F^2\lambda^2_{TF})^{3/2}} \tag{62}$$

Mobility expression:

$$\mu^{3d}_{disl.} = \frac{e}{m^*}\tau^{3d}_{disl.} = \frac{e}{m^*}\frac{m^* c_0^2}{n_{disl.} f^2_{disl.}\hbar}\frac{(3\,n_{3d})^{2/3}}{\pi^{8/3}}(1+4\,k_F^2\lambda^2_{TF})^{3/2}$$

$$k_F = \left(3\pi^2 n_{3d}\right)^{\frac{1}{3}} \tag{63}$$

$$\frac{1}{q_{TF}} = \text{Thomas-Fermi screening length, } \lambda_{TF} = \frac{\hbar k_F}{e}\sqrt{\frac{\epsilon_s}{3 n_{3d} m^*}} \tag{64}$$

$$\implies \lambda^4_{TF} = \frac{\hbar^4 k_F^4 \epsilon_s^2}{9\,e^4\,n_{3d}^2 m^{*2}} = \frac{3^{2/3}}{\pi^{8/3}}\frac{e}{\hbar}\frac{c_0^2\,n_{3d}^{2/3}}{n_{disl.} f^2_{disl.}}(1+4\,k_F^2\lambda^2_{TF})^{3/2}$$

$$4\,k_F^2\lambda^2_{TF} = \frac{4}{3}\frac{\hbar^2\,k_F^4\epsilon_s}{e^2\,n_{3d}\,m^*} = \frac{4\pi^{\frac{8}{3}}}{3}\frac{\hbar^2\,\epsilon_s\,(3\,n_{3d})^{4/3}}{e^2\,n_{3d}\,m^*} = 4\pi^{\frac{8}{3}}\frac{(3\,n_{3d})^{1/3}\,\hbar^2\,\epsilon_s}{e^2\,m^*} \tag{65}$$

Re-writing as function position, z:

$$\mu^{3d}_{disl.} = \frac{3^{2/3}}{\pi^{8/3}}\frac{e}{\hbar}\frac{c_0(z)^2\,n_{3d}(z)^{2/3}}{n_{disl.} f^2_{disl.}}(1+4\,k_F^2\lambda^2_{TF})^{3/2} \tag{66}$$

$$4\,k_F^2\lambda^2_{TF} = 4\pi^{8/3}\times 3^{\frac{1}{3}}\times\frac{\hbar^2}{e^2}\times\frac{n_{3d}(z)^{1/3}\,\epsilon_s(z)}{m^*(z)} \tag{67}$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\mu^{3d}_{disl.} = \frac{3^{2/3}}{\pi^{8/3}}\frac{e}{\hbar}\frac{c_0(z)^2\,n_{3d}(z)^{2/3}}{n_{disl.} f^2_{disl.}}(1+4\,k_F^2\lambda^2_{TF})^{3/2}$$

$$= 149.273\times\frac{c_0'(z)^2\,n_{3d}'(z)^{2/3}}{n'_{disl.} f'^2_{disl.}}(1+4\,k_F^2\lambda^2_{TF})^{3/2}\ [\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}]$$

$$4\,k_F^2\lambda^2_{TF} = 4\pi^{8/3}\times 3^{\frac{1}{3}}\times\frac{\hbar^2\epsilon_0}{e^2\,m_0}\times\frac{n_{3d}(z)^{1/3}\,\epsilon_s'(z)}{m^{*\prime}(z)}$$

$$= 0.0514\times\frac{n_{3d}'(z)^{1/3}\,\epsilon_s'(z)}{m^{*\prime}(z)} \tag{68}$$

Classical-quantum scattering time/rate ratio for charged dislocation scattering: The implementation is based on Refs.[11, 7, 6].

$$\frac{1}{\tau^q_{disl.}}(k) = \frac{m^*\,e^4\,n_{disl.} f^2_{disl.}}{c_0^2\,\epsilon_s^2\,\hbar^3}\frac{\lambda^4}{(1+4\,k^2\lambda^2)^{\frac{3}{2}}}(1+2\,k^2\lambda^2) \tag{69}$$

$$\frac{1}{\tau^c_{disl.}}(k) = \frac{m^*\,e^4\,n_{disl.}\,f^2_{disl.}}{c_0^2\,\epsilon_s^2\,\hbar^3}\frac{\lambda^4}{(1+4\,k^2\lambda^2)^{\frac{3}{2}}} \tag{70}$$

$$\implies \left.\frac{\tau^c_{disl.}}{\tau^q_{disl.}}\right|_D = 1+2\,k^2\lambda^2 \Rightarrow 1+2\,k_F^2\lambda^2_{TF} = 1+0.0257\times\frac{n_{3d}'^{1/3}\,\epsilon_s'}{m^{*\prime}}$$

5.1.2.4 Non-degenerate limit

- $\eta_F << -1$; $2\,k_t >> q_s$; $k_B T \geq \hbar\,\omega_{pl}$
- $q_s|_{ND} \equiv$ Debye screening wave-vector, $q_D = \sqrt{\frac{e^2\,n_{3d}}{\epsilon_s\,k_B T}}$

Scattering rate: from Eqn.49;

$$\begin{aligned}
\left.\frac{1}{\tau^c_{disl.}}\right|_{ND} &= \frac{n_{disl.}\,m^*\,e^4\,f^2_{disl.}}{\epsilon_s^2\,\hbar^3\,c_0^2}\,\frac{1}{q_D^4\left(\frac{4k^2}{q_D^2}\right)^{\frac{3}{2}}} \quad [\because 2\,k_t >> q_s] && (71)\\
&= \frac{n_{disl.}\,m^*\,e^4\,f^2_{disl.}}{\epsilon_s^2\,\hbar^3\,c_0^2}\,\frac{1}{q_D\,(2k)^3}\\
&= \frac{n_{disl.}\,m^*\,e^4\,f^2_{disl.}}{\epsilon_s^2\,c_0^2}\,\frac{(\epsilon_s k_B T)^{\frac{1}{2}}}{(e^2\,n_{3d})^{\frac{1}{2}}}\,\frac{1}{(2\hbar k)^3}\\
&= \frac{n_{disl.}\,m^*\,e^3\,f^2_{disl.}}{\epsilon_s^{\frac{3}{2}}\,c_0^2}\,\frac{(k_B T)^{\frac{1}{2}}}{(n_{3d})^{\frac{1}{2}}}\,\frac{1}{\left(2\sqrt{2m^* E_k}\right)^3} \quad \left[\because E_k = \frac{\hbar^2 k^2}{2m^*}\right]\\
&= \frac{n_{disl.}\,m^*\,e^3\,f^2_{disl.}}{\epsilon_s^{\frac{3}{2}}\,c_0^2}\,\frac{(k_B T)^{\frac{1}{2}}}{(n_{3d})^{\frac{1}{2}}}\,\frac{1}{(2\sqrt{2})^3\,(m^* E_k)^{\frac{3}{2}}}\\
&= \frac{n_{disl.}\,e^3\,f^2_{disl.}}{(2\sqrt{2})^3\,m^{*\frac{1}{2}}\,\epsilon_s^{\frac{3}{2}}\,c_0^2}\,\frac{(k_B T)^{\frac{1}{2}}}{(n_{3d})^{\frac{1}{2}}}\,\frac{1}{(k_B T)^{\frac{3}{2}}\left(\frac{E_k}{k_B T}\right)^{\frac{3}{2}}}\\
&= \frac{n_{disl.}\,e^3\,f^2_{disl.}}{(2\sqrt{2})^3\,m^{*\frac{1}{2}}\,\epsilon_s^{\frac{3}{2}}\,c_0^2\,k_B T\,n_{3d}^{\frac{1}{2}}}\left(\frac{E_k}{k_B T}\right)^{-\frac{3}{2}} && (72)\\
\implies \tau^c_{disl.}|_{ND} &\equiv \tau_0\left(\frac{E_k}{k_B T}\right)^{\frac{3}{2}} \quad \left[\tau_0 \equiv \frac{(2\sqrt{2})^3\,m^{*\frac{1}{2}}\,\epsilon_s^{\frac{3}{2}}\,c_0^2\,k_B T\,n_{3d}^{\frac{1}{2}}}{n_{disl.}\,e^3\,f^2_{disl.}}\right] && (73)
\end{aligned}$$

Mobility expression: In $3d$, following Eqn. 23.31 in Ref.[11]:

$$\begin{aligned}
\mu^{3d}_{disl.}|_{ND} &= \frac{e\tau_0}{m^*}\frac{\Gamma(\frac{5}{2}+r)}{\Gamma(\frac{5}{2})} && (74)\\
&= \frac{4}{3\sqrt{\pi}}\frac{e}{m^*}\frac{(2\sqrt{2})^3\,m^{*\frac{1}{2}}\,\epsilon_s^{\frac{3}{2}}\,c_0^2\,k_B T\,n_{3d}^{\frac{1}{2}}}{n_{disl.}\,e^3\,f^2_{disl.}} \times \Gamma\left(\frac{5}{2}+\frac{3}{2}\right) \quad \left[\because r = \frac{3}{2}\right]\\
&= \frac{4\times(3\times 2)\times(2\sqrt{2})^3}{3\sqrt{\pi}}\,\frac{\epsilon_s^{\frac{3}{2}}\,c_0^2\,k_B T\,n_{3d}^{\frac{1}{2}}}{m^{*\frac{1}{2}}\,n_{disl.}\,e^2\,f^2_{disl.}}\\
&= \frac{2^7\sqrt{2}}{\sqrt{\pi}}\,\frac{\epsilon_s^{\frac{3}{2}}\,c_0^2\,k_B T\,n_{3d}^{\frac{1}{2}}}{m^{*\frac{1}{2}}\,n_{disl.}\,e^2\,f^2_{disl.}} = \frac{128\sqrt{2}}{\sqrt{\pi}}\,\frac{\epsilon_s^{\frac{3}{2}}\,c_0^2\,k_B T\,n_{3d}^{\frac{1}{2}}}{m^{*\frac{1}{2}}\,n_{disl.}\,e^2\,f^2_{disl.}} && (75)
\end{aligned}$$

Re-writing as function position, z:

$$\mu^{3d}_{disl.}|_{ND} = \frac{128\sqrt{2}}{\sqrt{\pi}}\,\frac{\epsilon_s(z)^{\frac{3}{2}}\,c_0(z)^2\,k_B T\,n_{3d}(z)^{\frac{1}{2}}}{m^*(z)^{\frac{1}{2}}\,n_{disl.}\,e^2\,f^2_{disl.}} \qquad (76)$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\mu^{3d}_{disl.}|_{ND} &= \frac{128\sqrt{2}}{\sqrt{\pi}}\,\frac{\epsilon_s(z)^{\frac{3}{2}}\,c_0(z)^2\,k_B T\,n_{3d}(z)^{\frac{1}{2}}}{m^*(z)^{\frac{1}{2}}\,n_{disl.}\,e^2\,f^2_{disl.}} \\
&= \frac{128\sqrt{2}}{\sqrt{\pi}}\,\frac{\epsilon_0^{\frac{3}{2}}k_B}{m_0^{\frac{1}{2}}e^2}\,\frac{\epsilon'_s(z)^{\frac{3}{2}}\,c_0(z)^2\,T'\,n_{3d}(z)^{\frac{1}{2}}}{m^{*\prime}(z)^{\frac{1}{2}}\,n_{disl.}\,f'^2_{disl.}} \\
&= 0.152\times\frac{c'_0(z)^2\,T'}{n'_{disl.}\,f'^2_{disl.}}\sqrt{\frac{\epsilon'_s(z)^3\,n'_{3d}(z)}{m^{*\prime}(z)}} \qquad (77)
\end{aligned}$$

Classical-quantum scattering time/rate ratio for charged dislocation scattering: In the Boltzmann limit:

$$\begin{aligned}
\left.\frac{\tau^c_{disl.}}{\tau^q_{disl.}}\right|_{ND} &= 1+\frac{2\,k_\perp^2}{q_D^2} = 1+\frac{4\,m^*k_BT}{\hbar^2q_D^2} \quad \left[\because \langle E_{k_\perp}\rangle = k_BT = \frac{\hbar^2k_\perp^2}{2m^*}\right] \\
&= 1+\frac{4\,\epsilon_s m^*(k_BT)^2}{\hbar^2\,e^2\,n_{3d}} \quad \left[\because q_D = \sqrt{\frac{e^2\,n_{3d}}{\epsilon_s\,k_BT}}\right] \\
&= 1+\frac{4\,\epsilon_0 m_0 k_B^2}{\hbar^2\,e^2}\,\frac{\epsilon'_s m^{*\prime}T'^2}{n_{3d}} \\
&= 1+2.15\times10^{-5}\times\frac{\epsilon'_s m^{*\prime}T'^2}{n'_{3d}} \qquad (78)
\end{aligned}$$

5.1.2.5 Scattering from strain field of threading dislocation

5.1.2.6 General scattering rate/mobility expression

The derivation is adapted from Ref.[13, 11].
Momentum relaxation scattering rate:

$$\frac{1}{\tau^{st.}_{disl.}(k)} = \frac{n_{dis.}m^*E_D^2b_e^2}{4\hbar^3}\left(\frac{1-2\gamma}{1-\gamma}\right)^2\frac{1}{k^2}\left(1-\frac{1}{\sqrt{1+\frac{4k^2}{q_s^2}}}\right) \qquad (79)$$

$$\begin{aligned}
\implies \frac{1}{\tau^{st.}_{disl.}(E_k)} &= \frac{n_{dis.}m^*E_D^2b_e^2}{4\hbar^3}\left(\frac{1-2\gamma}{1-\gamma}\right)^2\frac{\hbar^2}{2m^*E_k}\left(1-\frac{1}{\sqrt{1+\frac{8m^*E_k}{\hbar^2q_s^2}}}\right) \quad \left[\because E_k = \frac{\hbar^2k^2}{2m^*}\right) \\
&= \frac{n_{dis.}E_D^2b_e^2}{8\hbar k_BT}\left(\frac{1-2\gamma}{1-\gamma}\right)^2\frac{1}{\varepsilon}\left(1-\frac{1}{\sqrt{\frac{8m^*k_BT}{h^2q_s^2}}\sqrt{\frac{h^2q_s^2}{8m^*k_BT}+\varepsilon}}\right) \qquad (80) \\
&= \frac{n_{dis.}E_D^2b_e^2}{8\hbar k_BT}\left(\frac{1-2\gamma}{1-\gamma}\right)^2\frac{1}{\varepsilon}\left(1-\frac{1}{\sqrt{1/B}\sqrt{B+\varepsilon}}\right) \\
\implies \tau^{st.}_{disl.}(E_k) &= A\varepsilon\left(1-\sqrt{\frac{B}{B+\varepsilon}}\right)^{-1} \qquad (81)
\end{aligned}$$

where $q_s = \sqrt{\frac{e^2N_c^{3d}F_{-\frac{1}{2}}(\eta_F)}{\epsilon_s\,k_BT}} = \sqrt{\frac{e^2n_{3d}}{\epsilon_s\,k_BT}}\sqrt{\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)}}$; $\varepsilon = \frac{E_k-E_c}{k_BT}$; $\eta_F = \frac{E_F-E_c}{k_BT}$;
$B = \frac{\hbar^2q_s^2}{8m^*\,k_BT} = \frac{\hbar^2e^2n_{3d}}{8m^*\epsilon_s\,(k_BT)^2}\,\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)}$; $A = \frac{8\hbar k_BT}{n_{dis.}E_D^2b_e^2}\left(\frac{1-\gamma}{1-2\gamma}\right)^2$

Mobility expression:

$$\mu^{st.}_{disl.} = -\frac{2e}{3m^*}\frac{\int_0^\infty \tau_c\,\varepsilon^{3/2}\,\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon}{\int_0^\infty f_0\times\varepsilon^{\frac{1}{2}}\,d\varepsilon} \tag{82}$$

$$= -\frac{2e}{3m^*}\frac{\int_0^\infty \tau_c\,\varepsilon^{3/2}\,\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon}{\Gamma(\frac{3}{2})\times F_{\frac{1}{2}}(\eta_F)}$$

$$= -\frac{4}{3\sqrt{\pi}}\frac{e}{m^*}\frac{\int_0^\infty \tau_c\,\varepsilon^{3/2}\,\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon}{F_{\frac{1}{2}}(\eta_F)}\left[\equiv -\frac{e}{m^*}\frac{\int_0^\infty \tau_c\,\varepsilon^{3/2}\,\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon}{\Gamma(\frac{5}{2})\times F_{\frac{1}{2}}(\eta_F)}\right] \tag{83}$$

$$= \frac{4}{3\sqrt{\pi}}\frac{e}{m^*}\frac{N_C^{3d}}{n_{3d}}\frac{AB}{2}\times I_{st.}(\eta_F)$$

$$= \frac{\sqrt{2}}{3\pi^2}\frac{e^3(k_BT)^{\frac{1}{2}}}{m^{*\frac{1}{2}}\epsilon_s n_{disl.}E_D^2 b_e^2}\left(\frac{1-\gamma}{1-2\gamma}\right)^2\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)}\times I_{st.}(\eta_F) \tag{84}$$

$$\int_0^\infty \tau_c\,\varepsilon^{3/2}\,\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon = \int_0^\infty A\varepsilon\left(1-\sqrt{\frac{B}{B+\varepsilon}}\right)^{-1}\times\varepsilon^{\frac{3}{2}}\times\frac{\partial f_0(\varepsilon)}{\partial\varepsilon}d\varepsilon$$

$$= A\left[-\int_0^\infty\frac{\partial}{\partial\varepsilon}\left[\varepsilon^{\frac{5}{2}}\left(1-\sqrt{\frac{B}{B+\varepsilon}}\right)^{-1}\right]f_0\,d\varepsilon\right]$$

$$= -\frac{AB}{2}\int_0^\infty\frac{5-5y-\frac{\varepsilon}{B+\varepsilon}y}{B(1-y)^2}\frac{\varepsilon^{\frac{3}{2}}}{1+e^{\varepsilon-\eta_F}}\,d\varepsilon \equiv -\frac{AB}{2}\times I_{st.}(\eta_F) \tag{85}$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$B = \frac{\hbar^2e^2n_{3d}}{8m^*\epsilon_s\,(k_BT)^2}\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)} = 23210.2\times\frac{n'_{3d}}{m^{*\prime}\epsilon'_sT'^2}\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)} \tag{86}$$

$$\mu^{st.}_{disl.} = \frac{\sqrt{2}}{3\pi^2}\frac{e^3(k_BT)^{\frac{1}{2}}}{m^{*\frac{1}{2}}\epsilon_s n_{disl.}E_D^2 b_e^2}\left(\frac{1-\gamma}{1-2\gamma}\right)^2\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)}\times I_{st.}(\eta_F)$$

$$= 3364750.021\times\frac{T'^{\frac{1}{2}}}{m^{*\prime\frac{1}{2}}\epsilon'_sn'_{disl.}E'^2_Da'^2_0}\left(\frac{1-\gamma}{1-2\gamma}\right)^2\frac{F_{-\frac{1}{2}}(\eta_F)}{F_{\frac{1}{2}}(\eta_F)}I_{st.}(\eta_F)\ \ [\mathrm{cm^2\,V^{-1}\,s^{-1}}] \tag{87}$$

$$I_{st.}(\eta_F) = \int_0^\infty\frac{5-5y-\frac{\varepsilon}{B+\varepsilon}y}{B(1-y)^2}\frac{\varepsilon^{\frac{3}{2}}}{1+e^{\varepsilon-\eta_F}}\,d\varepsilon\ ;y=\sqrt{\frac{B}{B+\varepsilon}} \tag{88}$$

• Isotropic Poisson's ratio, $\gamma = \frac{C_{12}}{C_{11}+C_{12}}$.
NOTE-1: For WZ structure the Poisson ratio is anisotropic[14]. However, the strain field due to edge dislocation uses 'isotropic/average Poisson ratio'. We use the definition $\gamma = C_{12}/(C_{11}+C_{12})$ following Ref.[15]. This was also used in the thesis, Ref.[3].
NOTE-2: In Ref.[14] there is a printing mistake. The s_{11} expression should be $2s_{11} = sc_{33} + \frac{1}{c_{11}-c_{12}}$.
• Magnitude of Burger's vector, $b_e = \frac{1}{3} < 11\overline{2}0 >$ for WZ structure, is approximated to be same as a_0.
• η_F is calculated using inverse Fermi-Dirac function (i.e., $\eta_F = F^{-1}_{1/2}(\frac{n_{3d}}{N_C^{3d}})$).
• $F_{-\frac{1}{2}}(\eta_F)$, $F_{\frac{1}{2}}(\eta_F)$ and $I_{st.}(\eta_F)$ are then calculated using numerical integration.

5.2 Phonon related scattering

5.2.1 Deformation potential scattering

The implementation is based on Refs.[3]. For deformation potential acoustic phonon mediated scattering $p = -\frac{1}{2}$. The **momentum relaxation time:**

$$\tau_{0,DP} = \frac{\hbar}{2\sqrt{\pi}}\frac{\rho\,v_{LA}^2}{E_D^2\,N_C^{3d}} \tag{89}$$

Mobility expression:

$$\begin{aligned}
\mu_{3d,DP} &= \frac{4e\,\tau_{0,DP}}{3\sqrt{\pi}\,m^*}\frac{1}{F_{\frac{1}{2}}(\eta_F)}\Gamma(2)\times F_0(\eta_F)\ \text{[From Eqn.16]} \\
&= \frac{4e\hbar}{6\pi\,m^*}\frac{\rho\,v_{LA}^2}{E_D^2}\frac{1}{N_C^{3d}\,F_{\frac{1}{2}}(\eta_F)}F_0(\eta_F) \\
&= \frac{2e\hbar}{3\pi\,m^*}\frac{\rho\,v_{LA}^2}{E_D^2\,n_{3d}}F_0(\eta_F)\ \text{[From Eqn.14]} \qquad (90)
\end{aligned}$$

Re-writing as function position, z:

$$\mu_{3d,DP}(z) = \frac{2e\hbar}{3\pi}\frac{\rho(z)\,v_{LA}(z)^2}{m^*(z)\,E_D(z)^2\,n_{3d}(z)}F_0(\eta_F(z)) \qquad (91)$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}
\mu_{3d,DP}(z) &= \frac{2e\hbar}{3\pi}\frac{\rho(z)\,v_{LA}(z)^2}{m^*(z)\,E_D(z)^2\,n_{3d}(z)}F_0(\eta_F(z)) \\
&= 1.5333\times10^{-6}\frac{\rho'(z)\,v'_{LA}(z)^2}{m^{*\prime}(z)\,E'_D(z)^2\,n'_{3d}(z)}F_0(\eta_F(z))\ [\text{cm}^2\,\text{V}^{-1}\,\text{s}^{-1}] \qquad (92)
\end{aligned}$$

5.2.2 Piezoelectric phonon scattering

The implementation is based on Refs.[3, 2]. Note following [2] we use K^2 term in our below derivation which is related to the terms in Ref.[3] as:

$$\rho v_s^2 \equiv C^*; K^2 \equiv \left\langle \frac{e_{pz}^2}{C^*\epsilon_s} \right\rangle = \left\langle \frac{e_{pz}^2}{\rho v_s^2\,\epsilon_s} \right\rangle$$

Here, following [16, 17], we calculate K^2 using the spherical averaging:

$$\begin{aligned}
K^2 &= \frac{\langle e_{LA}^2\rangle}{\epsilon_s\,\rho\,v_{LA}^2} + \frac{\langle e_{TA}^2\rangle}{\epsilon_s\,\rho\,v_{TA}^2} \qquad (93) \\
\langle e_{LA}^2\rangle &= \frac{1}{105}\left[15\,e_{33}^2 + 12e_{33}(e_{31}+2e_{15}) + 8(e_{31}+2e_{15})^2\right] \qquad (94) \\
\langle e_{TA}^2\rangle &= \frac{1}{105}\left[48\,e_{15}^2 + 16e_{15}(e_{33}-e_{15}-e_{31}) + 6(e_{33}-e_{15}-e_{31})^2\right] \qquad (95)
\end{aligned}$$

For piezoelectric phonon scattering $p = +\frac{1}{2}$. The **momentum relaxation time:**

$$\tau_{0,PE} = \frac{2m^*\epsilon_s\,k_BT}{\sqrt{\pi}\,\hbar\,e^2\,K^2}\frac{1}{N_C^{3d}} \qquad (96)$$

Mobility expression:

$$\begin{aligned}
\mu_{3d,PE} &= \frac{4e\,\tau_{0,PE}}{3\sqrt{\pi}\,m^*}\frac{1}{F_{\frac{1}{2}}(\eta_F)}\Gamma(3)\times F_1(\eta_F)\ \text{[From Eqn.16]} \\
&= \frac{16\epsilon_s\,k_BT}{3\pi\hbar e K^2}\frac{1}{N_C^{3d}\,F_{\frac{1}{2}}(\eta_F)}F_1(\eta_F) \\
&= \frac{16\,k_BT}{3\pi\hbar e}\frac{\epsilon_s}{K^2\,n_{3d}}F_1(\eta_F)\ \text{[From Eqn.14]} \qquad (97)
\end{aligned}$$

Re-writing as function position, z:

$$\mu_{3d,PE}(z) = \frac{16\,k_BT}{3\pi\hbar e}\frac{\epsilon_s(z)}{K^2(z)\,n_{3d}(z)}F_1(\eta_F(z)) \qquad (98)$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}\mu_{3d,PE}(z) &= \frac{16\,k_B}{3\pi\hbar e}\frac{\epsilon_s(z)\,T}{K^2(z)\,n_{3d}(z)}F_1(\eta_F(z)) \\ &= 0.12283\times\frac{\epsilon_s'(z)\,T'}{K^{2'}(z)\,n_{3d}'(z)}F_1(\eta_F(z))\ [\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}] \end{aligned} \tag{99}$$

5.2.3 Polar optical phonon scattering

The implementation is based on Refs.[11, 3, 4]. Note that, since POP scatterings are inelastic in nature and there is no simple energy dependence unlike mechanisms discussed above, there is no general analytical expression for the POP carrier mobility. However, assuming POP energies satisfy $\hbar\omega_{LO} >> k_BT$ (results from BTE when assuming a much higher emission rate than the absorption rate) the **mobility** is:

$$\begin{aligned}\mu_{3d,POP} &= \frac{4\pi\epsilon^*\hbar}{e\,m^*q_0}\frac{1}{N_{BE}} \\ &= \frac{2^{\frac{3}{2}}\pi\epsilon^*\hbar^2}{e\,m^{*\frac{3}{2}}E_{pop}^{\frac{1}{2}}}\times\left(e^{\frac{E_{pop}}{k_BT}}-1\right) \end{aligned} \tag{100}$$

$$\begin{aligned}\frac{1}{\epsilon^*} &= \frac{1}{\epsilon_h}-\frac{1}{\epsilon_s} \\ \hbar\omega_{pop} &= \frac{\hbar^2\,q_0^2}{2m^*} \implies q_0=\sqrt{\frac{2m^*\,E_{POP}}{\hbar^2}} \\ N_{BE} &= \frac{1}{e^{\frac{\hbar\omega_{pop}}{k_BT}}-1}\ [\text{Bose-Einstein distribution}] \end{aligned} \tag{101}$$

Re-writing as function position, z:

$$\mu_{3d,POP}(z) = \frac{2^{\frac{3}{2}}\pi\hbar^2}{e}\frac{\epsilon^*(z)}{m^*(z)^{\frac{3}{2}}E_{pop}(z)^{\frac{1}{2}}}\times\left(e^{\frac{E_{pop}(z)}{k_BT}}-1\right) \tag{102}$$

Units (with 'prime' are unitless numbers; using units from parameter table from Sec.1):

$$\begin{aligned}\mu_{3d,POP}(z) &= \frac{2^{\frac{3}{2}}\pi\hbar^2}{e}\frac{\epsilon^*(z)}{m^*(z)^{\frac{3}{2}}E_{pop}(z)^{\frac{1}{2}}}\times\left(e^{\frac{E_{pop}(z)}{k_BT}}-1\right) \\ &= 0.15693\times\frac{\epsilon^{*'}(z)}{m^{*'}(z)^{\frac{3}{2}}E_{pop}'(z)^{\frac{1}{2}}}\left(e^{11604.52\times\frac{E_{pop}'(z)}{T'}}-1\right)\ [\mathrm{cm}^2\,\mathrm{V}^{-1}\,\mathrm{s}^{-1}] \end{aligned} \tag{103}$$

6 General 3DEG properties

1. Fermi energy w.r.t k_BT (parabolic dispersion)

$$\eta_F = F_{\frac{1}{2}}^{-1}\left(\frac{n_{3d}}{N_c^{3d}}\right) \tag{104}$$

$$\eta_F|_D = \frac{E_F}{k_BT} = \frac{\hbar^2\,k_F^2}{2\,m^*\,k_BT} \tag{105}$$

2. Fermi energy (parabolic dispersion)

$$E_F = \eta_F\times k_BT = 8.62\times10^{-5}\times\eta_F'\times T'\ [\mathrm{eV}] \tag{106}$$

$$\begin{aligned}E_F|_D &= \frac{\hbar^2\,k_F^2}{2\,m^*} = \frac{\hbar^2}{2\,m_0}\frac{k_F'^2}{m^{*'}}\times(10^6\,\mathrm{cm}^{-1})^2 \\ &= \frac{\hbar^2}{2\,m_0\,e'}10^{12}\times\frac{k_F'^2}{m^{*'}} = 0.00038\times\frac{k_F'^2}{m^{*'}}\ [\mathrm{eV}] \end{aligned} \tag{107}$$

3. Screening wave-vector

$$
\begin{aligned}
q_s &= \sqrt{\frac{e^2 N_c^{3d} F_{-\frac{1}{2}}(\eta_F)}{\epsilon_s\, k_B T}} = \sqrt{\frac{e^2}{\epsilon_s\, k_B T} 2 \left(\frac{m^* k_B T}{2\pi\hbar^2}\right)^{\frac{3}{2}} F_{-\frac{1}{2}}(\eta_F)} \\
&= \sqrt{\frac{e^2 m^{*\frac{3}{2}} (k_B T)^{\frac{1}{2}}}{\epsilon_s (2\pi^3)^{\frac{1}{2}} \hbar^3} F_{-\frac{1}{2}}(\eta_F)} \\
&= \sqrt{\frac{e^2 m_0 \frac{3}{2} k_B^{\frac{1}{2}}}{\epsilon_0 (2\pi^3)^{\frac{1}{2}} \hbar^3}} \sqrt{\frac{m^{*\prime\frac{3}{2}} T'^{\frac{1}{2}}}{\epsilon'_s} F_{-\frac{1}{2}}(\eta_F)} \\
&= 10.07 \times \sqrt{\frac{m^{*\prime\frac{3}{2}} T'^{\frac{1}{2}}}{\epsilon'_s} F_{-\frac{1}{2}}(\eta_F)} \;\; [10^6\,\mathrm{cm}^{-1}] \qquad (108)
\end{aligned}
$$

$$
\begin{aligned}
q_s|_{ND} \equiv q_D &= \sqrt{\frac{e^2\, n_{3d}}{\epsilon_s\, k_B T}} = \sqrt{\frac{e^2}{\epsilon_0\, k_B}} \sqrt{\frac{n_{3d}}{\epsilon'_s T'}} \\
&= 144.91 \times \sqrt{\frac{n'_{3d}}{\epsilon'_s T'}} \;\; [10^6\,\mathrm{cm}^{-1}] \qquad (109)
\end{aligned}
$$

$$
\begin{aligned}
q_s|_D \equiv q_{TF} &= 1/\lambda_{TF} = \frac{e}{\hbar k_F} \sqrt{\frac{3 n_{3d} m^*}{\epsilon_s}} = \frac{e}{\hbar\,(3\pi^2 n_{3d})^{\frac{1}{3}}} \sqrt{\frac{3 n_{3d} m^*}{\epsilon_s}} \\
&= \frac{e\, m_0^{\frac{1}{2}}\, 3^{\frac{1}{6}}}{\hbar\, \epsilon_0^{\frac{1}{2}} \pi^{\frac{2}{3}}} \frac{m^{*\prime\frac{1}{2}} n_{3d}^{\prime\frac{1}{6}}}{\epsilon_s^{\prime\frac{1}{2}}} \times 10^3\,\mathrm{cm}^{-1} \\
&= 27.28 \times \sqrt{\frac{m^{*\prime} n_{3d}^{\prime\frac{1}{3}}}{\epsilon_s}} \;\; [10^6\,\mathrm{cm}^{-1}] \qquad (110)
\end{aligned}
$$

4. Classical-quantum scattering time/rate ratio for charged dislocation

$$
\left.\frac{\tau^c_{dis}}{\tau^q_{dis}}\right|_D = 1 + 2\, k_F^2 \lambda_{TF}^2 = 1 + 0.0257 \times \frac{n'^{1/3}_{3d}\, \epsilon'_s}{m^{*\prime}} \qquad (111)
$$

$$
\left.\frac{\tau^c_{dis}}{\tau^q_{dis}}\right|_{ND} = 1 + 2.15 \times 10^{-5} \times \frac{\epsilon'_s m^{*\prime} T'^2}{n'_{3d}} \qquad (112)
$$

5. Fermi wave vector

$$
k_F = \left(3\pi^2 n_{3d}\right)^{\frac{1}{3}} = 3.09 \times n_{3d}^{\prime\frac{1}{3}} \;\; [10^6\,\mathrm{cm}^{-1}] \qquad (113)
$$

6. POP wave vector

$$
\begin{aligned}
q_0 &= \sqrt{\frac{2 m^*\, E_{POP}}{\hbar^2}} = \sqrt{\frac{2\, m_0\, e'}{\hbar^2}} \sqrt{m^{*\prime}\, E'_{POP}} \\
&= 51.23 \times \sqrt{m^{*\prime}\, E'_{POP}} \;\; [10^6\,\mathrm{cm}^{-1}] \qquad (114)
\end{aligned}
$$

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